AmodalEdit: Advancing Instruction-Based Image Editing via Amodal Scene Layout Perception

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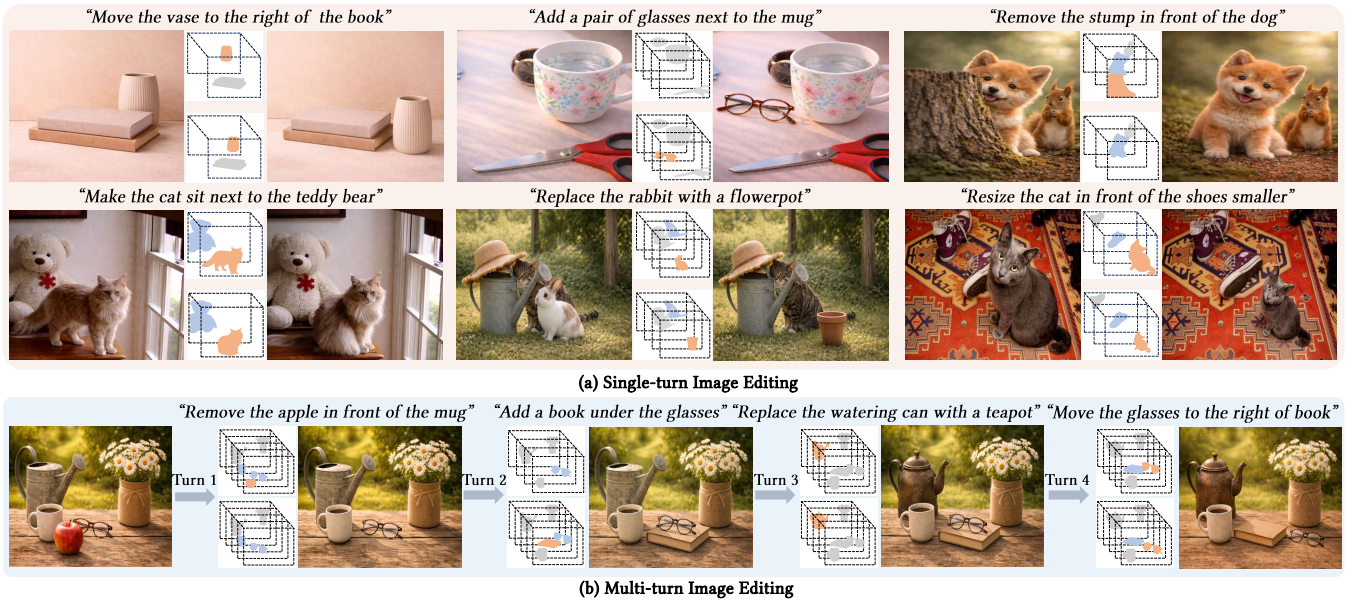


Figure 1: Occlusion-aware image editing. For each case (a), given an input image with the text instruction (left), *AmodalEdit* automatically predicts amodal scene layouts before and after the instruction (middle), encoding target objects (orange) and occlusion-related objects (blue) explicitly. These priors are leveraged to generate the edited image (right) by guiding target objects following instructions and preserving the integrity of non-target objects. We further show the superiority of *AmodalEdit* in multi-turn image editing (b), where geometric and semantic consistency across editing sequences could be maintained.

Abstract

Instruction-based image editing has recently attracted substantial interest for generative multimedia. While significant advances have been achieved, existing works mainly rely on the explicit visibility of target objects, posing significant challenges of geometric inconsistencies and texture artifacts when editing partially occluded objects. We demonstrate that such an assumption is often incorrect, as object interactions and occlusions commonly exist in scenarios. To tackle this challenge, we propose AmodalEdit, a new framework for instruction-based image editing via amodal scene layout perception. In particular, given an input image with text instructions, we first introduce an instruction graph construction module that maps the input modalities into a structured representation with object editing and occlusion attributes. We then adopt an amodal layout perception module to infer complete object masks and underlying relationships. Finally, we propose an instruction

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amodal adapter module to enforce global scene coherence by guiding target objects with instructions, while preserving the physical integrity of non-target objects. Experimental results demonstrate that AmodalEdit consistently surpasses state-of-the-art methods in structural integrity and physical plausibility. Furthermore, we showcase the superiority of AmodalEdit in multi-turn image editing by maintaining the amodal scene layout across editing sequences.

CCS Concepts

• **Computing methodologies** → **Artificial intelligence**.

Keywords

Multimodal Fusion, Image Editing, Scene Understanding

ACM Reference Format:

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1 Introduction

Instruction-based image editing [4, 6, 32, 35, 37, 41] has attracted substantial interest by manipulating visual content with user instructions. With the rapid advancements in generative models [10, 11, 38], recent methods have demonstrated capabilities in object manipulation, style transfer, and attribute editing. Among different interaction paradigms, text instructions [3, 13–15, 22, 40, 48, 51, 54, 55] provide an intuitive way for non-expert users in multimedia applications to manipulate images with natural language descriptions.

Despite remarkable progress with local or global editing operations, existing works [3, 48] mainly rely on the visible appearance of target objects, implicitly assuming that target objects are fully exposed and unobstructed in the scene. We argue that such an assumption oversimplifies user intention, as object occlusions commonly exist in real-world scenarios. When a user attempts to edit a partially occluded object, we need to maintain the scene structure and ensure the partially occluded object naturally blends in the edited image. Without fine-grained scene understanding abilities, existing methods inevitably suffer from geometric inconsistencies and texture artifacts in various scenarios with object occlusions.

Recall how the human visual system works in these scenarios. Humans would first infer the potential geometric properties (e.g., shape, pose) and semantic relations (e.g., support, containment) of different objects in the scene. As illustrated in Figure 1, the geometric properties contain common object attributes (e.g., the shape of the vase), while the semantic relations represent spatial and sequential arrangement among objects (e.g., relations between the vase and the book). Such a cognitive ability allows humans to modify target objects while implicitly following physical laws and maintaining coherence of the entire scene.

Inspired by the above observation, we propose *AmodalEdit*, a new framework for instruction-based image editing via amodal scene layout perception. Different from existing works that rely on pixel-wise attention to enforce text-image manipulation consistency, we aim to explore object-wise scene reasoning to infer the geometry and semantic properties of target and non-target objects. In particular, given an input image with text instructions, we first introduce an Instruction Graph Construction (IGC) module that maps the input modalities into a structured representation of editing targets and occlusion hierarchies. We then adopt an Amodal Layout Perception (ALP) module to infer complete object masks and underlying spatial-depth object relationships. Finally, we propose an Instruction Amodal Adapter (IAA) module to enforce scene coherence by guiding editable objects with instructions, while preserving the physical integrity of non-target objects.

To thoroughly evaluate the effectiveness of our method, we construct OIE-Bench (**O**cclusion-aware **I**mage **E**diting **Bench**mark), a new benchmark with 213 samples, featuring object interactions and editing operations in diverse scenarios. Each case consists of a source image, a text prompt, and a target image. Experimental results demonstrate that *AmodalEdit* outperforms the state-of-the-art methods both qualitatively and quantitatively. We further show the superiority of *AmodalEdit* in multi-turn image editing by maintaining the amodal scene layout consistency across editing sequences.

Our main contributions are summarized as follows:

- We make the first attempt to address the object occlusion issues for instruction-based image editing, advancing coherent manipulation of object interactions across diverse scenarios.
- We propose *AmodalEdit*, a novel framework for instruction-based image editing via amodal scene layout perception, enforcing target objects to follow instructions and preserving the integrity of non-target objects.
- Extensive evaluations show that *AmodalEdit* outperforms state-of-the-art methods in structural integrity and physical plausibility, and improves multi-turn editing performance.

2 Related Work

Instruction-based Image Editing. Instruction-based image editing allows for efficient manipulation with user intention guidance. Early works [18] utilize generative adversarial networks for image manipulation in a restricted latent space. Diffusion models [36, 39] have significantly improved editing effectiveness by supporting text instructions. InstructPix2Pix [3] leverages large language models [42] to generate synthetic image-instruction triplets and trains a diffusion model to follow text instructions. MagicBrush [50] utilizes human-annotated data to improve faithfulness, while UltraEdit [54] leverages large-scale instruction training for fine-grained image manipulation. MGIE [7] incorporates multimodal large language models (MLLMs) to derive expressive instructions. To improve spatial editing accuracy, SmartEdit [13], PRG-Edit [15], and SuperEdit [22] refine spatial conditioning and guidance mechanisms, while FireEdit [55] explicitly models localized modifications to prevent unintended background alterations. IEAP [12] formulates instruction-based image editing as compositions of atomic operations to better handle structurally inconsistent edits. Concurrently, efforts have shifted toward building unified, scalable frameworks capable of handling diverse editing operations. EditAR [30] formulates editing as an autoregressive conditional generation problem. ICEdit [53] leverages large-scale Diffusion Transformers (DiTs) with in-context conditioning. Step1X-Edit [28] integrates multimodal reasoning with diffusion generation.

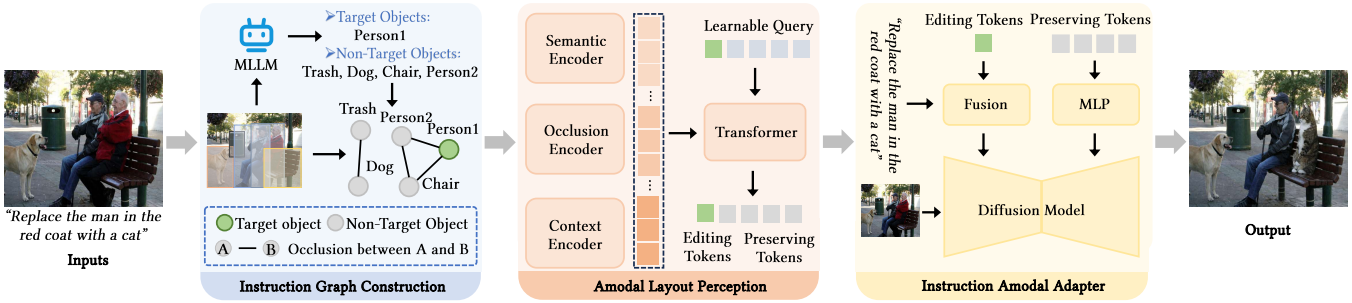


Figure 2: The overall pipeline of *AmodalEdit* for instruction-based image editing. Given an input image and the text instruction, the Instruction Graph Construction (IGC) module first maps the input modalities into a structured representation with object editing and occlusion attributes. The Amodal Layout Perception (ALP) module then infers complete object masks and underlying object relationships. Finally, the Instruction Amodal Adapter (IAA) module enforces global scene coherence by guiding editable objects with instructions, while preserving the physical integrity of non-target objects.

While remarkable advances have been achieved, these methods mainly rely on the explicit visibility of target objects, posing significant challenges of geometric inconsistencies and texture artifacts when editing occluded objects. We demonstrate that such an assumption is often incorrect, as object interactions and occlusions commonly exist in real-world scenarios. Although SGEdit [52] considers object interactions for image editing via scene graph prediction, it mainly focuses on object-centric edits and explicitly represented inter-object relations, while changes in occlusion relationships are not explicitly considered. In contrast, our approach advances instruction-based image editing performance by addressing object occlusion issues through amodal scene layout perception.

Amodal Segmentation. Amodal segmentation [21] aims to recover the shape of partially occluded objects. Early works [5, 49] formulate it as an extension of instance segmentation by jointly predicting visible and amodal masks. To address the ambiguity of heavy occlusions, representative approaches [24, 47] learn canonical or generative shape representations from partial observations. Other methods incorporate geometric cues, e.g., 3D reconstruction guidance [23], or hierarchical recovery strategies [8] that estimate global structure before refining local details. Diffusion-based methods [43] provide strong priors for plausible occluded structure generation. Additional priors have also been considered [25, 31], including occlusion relationship reasoning, uncertainty-aware boundary modeling, and multi-view relational learning. Recent efforts [1, 26, 29] further extend amodal segmentation to zero-shot settings.

However, these methods are primarily designed for scene segmentation and lack the mechanism to adopt amodal masks into editing priors. Unlike prior works, our method focuses on utilizing amodal perception priors for instruction-based image editing, allowing our method to exhibit remarkable robustness to challenging single-turn and multi-turn editing operations.

3 Method

3.1 Overview

Our goal is to advance text instruction-based image editing in diverse scenarios with object occlusions. Existing works [7, 41, 48, 54] mainly rely on the explicit visibility of target objects. In contrast, we present *AmodalEdit*, a novel framework for instruction-guided image editing that offers robustness to both single-turn and multi-turn image editing operations. As illustrated in Figure 2, the editing workflow consists of three core components, i.e., the Instruction Graph Construction (IGC) module, the Amodal Layout Perception (ALP) module, and the Instruction Amodal Adapter (IAA) module.

Given an input image and an instruction, IGC first adopts an MLLM to map the input modalities into a structured representation with object editing and occlusion attributes. Based on the obtained representation, ALP utilizes semantic features, occlusion features, and context features to infer complete object masks and underlying object relationships. Finally, IAA injects the perceived amodal scene layout priors into the diffusion-based model to enforce global scene coherence by guiding editable objects with instructions, while preserving the physical integrity of non-target objects.

3.2 Instruction Graph Construction

Our pipeline begins by constructing an instruction graph from the input image and instruction. The objective of IGC is to map the inputs into a structured representation with object editing and occlusion attributes. Unlike traditional scene graphs with spatial relationships, IGC is designed to capture the dynamic editing intention while preserving the latent occlusion hierarchy of the scene.

For each object in the input image, we employ an off-the-shelf object detector [20] to extract its crucial attributes, including bounding box, category, and visual appearance. We then extract the target and non-target objects in the scene through interactions with the MLLM. Detailed prompt templates are designed carefully to guide responses of the MLLM, ensuring its outputs contain both object editing and occlusion information accurately. In particular, the MLLM separates the objects into two mutually exclusive subsets: a target object set and a non-target object set. The target object set contains instances directly targeted by the instruction, such as those undergoing changes in geometry, appearance, or visibility. The non-target object set contains background or instances that should remain structurally intact or be plausibly completed if newly exposed. In addition, the occlusion relationships between objects in the scene are derived as the edge representation, serving as a structured prior for subsequent amodal reasoning.

3.3 Amodal Layout Perception

Given the scene understanding priors from the above structured representation, ALP aims to infer complete and occlusion-aware representations for all objects in the scene. For each object i with bounding box b_i, we first extract its complementary features through three specific encoders as follows.

Semantic Encoder. We crop the object region based on its bounding box b_i and feed it into a pretrained CLIP image encoder to obtain the visual feature f_i^v. The object category is fed into the CLIP text encoder to obtain the semantic feature f_i^t. We compute a gate vector g_i to balance the visual and semantic features adaptively:

$$g_i = \sigma\left(\mathrm{MLP}([f_i^v; f_i^t])\right), \tag{1}$$

where $[\cdot;\cdot]$ denotes feature concatenation and $\sigma(\cdot)$ is the sigmoid function. The semantic descriptor sd_i is obtained as:

$$sd_i = g_i \odot f_i^v + (1 - g_i) \odot f_i^t, \tag{2}$$

where $\odot$ denotes element-wise multiplication. Such a formulation enables the model to emphasize the visual feature when the visible appearance is reliable, while relying more on the category prior when the visual evidence is ambiguous.

To incorporate spatial information, we normalize the bounding box coordinates and encode them with a lightweight geometric encoder $\phi_b(\cdot)$. The semantic and geometric descriptors are then concatenated and projected into the latent space:

$$\tilde{S}_i = \mathrm{MLP}([sd_i; \phi_b(b_i)]). \tag{3}$$

We further add a learnable object-role embedding r^{obj} to indicate that the token corresponds to the object itself. The resulting semantic token S_i serves as the primary semantic anchor in subsequent amodal reasoning:

$$S_i = \tilde{S}_i + r^{obj}. \tag{4}$$

Occlusion Encoder. For each object i, its candidate occluders are determined according to the directed occlusion relations from IGC. Specifically, object j is treated as an occluder of object i if the instruction graph contains an edge $j \to i$. When multiple occluders exist, we rank them according to the occlusion sizes with respect to object i and retain the top K instances based on a threshold.

Let (c_i^x, c_i^y, w_i, h_i) and (c_j^x, c_j^y, w_j, h_j) denote the center coordinates, width, and height of bounding boxes b_i and b_j, respectively. For each occluder j, we define its relative geometric descriptor r_{ij} with respect to object i, capturing the relative center offsets and scale differences between the occluder and the target object:

$$r_{ij} = \left[\frac{c_j^x - c_i^x}{w_i}, \frac{c_j^y - c_i^y}{h_i}, \log\frac{w_j}{w_i}, \log\frac{h_j}{h_i}\right]. \tag{5}$$

The relative geometry is encoded with a lightweight encoder as:

$$e_{ij}^r = \phi_r(r_{ij}). \tag{6}$$

We then concatenate e_{ij}^r with the semantic descriptor sd_j of the occluding object and project it into the latent space:

$$\tilde{O}_{ij} = \mathrm{MLP}([sd_j; e_{ij}^r]). \tag{7}$$

We also add a learnable occluder-role embedding r^{occ} to obtain the occlusion token associated with occluder j. The occlusion tokens of object i are collected as $O_i = O_{ij} \mid j \to i$:

$$O_{ij} = \tilde{O}_{ij} + r^{occ}. \tag{8}$$

Different from the object-role embedding r^{obj}, the occluder-role embedding r^{occ} indicates that the token originates from an external instance providing occlusion context. It enables the decoder to differentiate the target object from surrounding occlusion cues, and reason effectively which neighboring objects are responsible for the invisible regions and how they affect the underlying structure.

Context Encoder. We extract scene contextual features from the input image using a hierarchical backbone paired with a feature pyramid network [27]. By applying RoIAlign [9] to b_i across multiple pyramid levels and flattening the aligned feature maps along the spatial dimensions, we obtain a sequence of patch-level context tokens C_i, which preserve local appearance around the visible region of the object and provide contextual cues for amodal prediction.

Finally, the outputs of these encoders are combined to form a memory bank $M_i = [S_i; O_i; C_i]$, where S_i encodes the semantic identity and coarse geometry of the target object, O_i summarizes explicit occlusion-aware contextual cues from relevant occluders, and C_i provides patch-level local visual evidence. We then employ a transformer-based decoder to infer the complete object structure from partial observations. Specifically, a set of learnable object queries is used to interact with M_i through cross-attention and iteratively refined to retrieve object-centric structural information. The final hidden states of the decoder are taken as the amodal object tokens T_i, which encode a holistic, occlusion-aware representation of the underlying object beyond its visible region. We further divide T_i into two groups based on the constructed instruction graph. One is the editing token set $\mathcal{T}_e$ that corresponds to objects in the target object set, and the other is the preserving token set $\mathcal{T}_p$ that corresponds to objects in the non-target object set.

To ensure that amodal object tokens T_i capture physically meaningful structure, we pass these tokens to an amodal mask decoder to predict the amodal mask $\hat{m}_i$. Note that the amodal mask decoder is only used for training, enforcing the learned tokens to represent the complete latent structure of the object, rather than merely memorizing visible appearance patterns.

3.4 Instruction Amodal Adapter

IAA aims to incorporate amodal priors to enforce global scene coherence in the edited image. We propose a dual-path injection mechanism to model target and non-target object cues separately, enabling precise manipulations on target objects, while preserving the physical integrity of non-target objects.

For non-target objects, we design a preserving pathway in which the intermediate features of the diffusion model query the preserving tokens $\mathcal{T}_p$ through a dedicated cross-attention layer. The obtained feature F_p serves as a soft structural prior that encourages the model to maintain the layout of non-target objects, while plausibly completing newly exposed regions caused by the instruction.

For target objects, we design an editing pathway to align the injected object priors with the instruction. Specifically, let Q_e denote a set of learnable editing queries and c_{txt} denote the instruction. We

first project the instruction embedding into the query space and use it to modulate the learnable queries:

$$\hat{Q}_e = Q_e + \phi_t(c_{\text{txt}}), \tag{9}$$

where $\phi_t(\cdot)$ denotes a learnable linear projection and $\hat{Q}_e$ denotes the instruction-conditioned editing queries. The fusion between the instruction-conditioned queries and the target-related amodal priors is then performed through cross-attention, where $\hat{Q}_e$ serves as the query, and $\mathcal{T}_e$ serves as the key and value. By doing so, the text instruction actively guides the retrieval of target-relevant structural information from the editing objects, ensuring that the injected amodal prior is aligned with the intended edit explicitly.

$$F_e = \text{CrossAttn}(\hat{Q}_e, \mathcal{T}_e, \mathcal{T}_e). \tag{10}$$

Finally, the outputs from the above two paths are aggregated to form the final conditioned feature Z' with a balancing coefficient λ, which controls the contribution of the preserving structure prior. Z' is injected into the diffusion model to guide the editing results:

$$Z' = F_e + \lambda F_p. \tag{11}$$

3.5 Training

Our model is trained with a two-stage paradigm. We first train the ALP module with an amodal dataset [56], containing paired visible and amodal masks. Given an input image, the ALP module predicts the amodal object tokens T_i, which are fed into a decoder to predict the amodal mask $\hat{m}_i$ and enforce object structural completeness. The training objective $\mathcal{L}_{\text{ALP}}$ is:

$$\mathcal{L}_{\text{ALP}} = \mathcal{L}_{\text{BCE}} + \lambda_m \mathcal{L}_{\text{Dice}}, \tag{12}$$

where $\mathcal{L}_{\text{BCE}}$ and $\mathcal{L}_{\text{Dice}}$ denote the weighted BCE loss and the Dice loss, respectively. λ_m is a balancing coefficient.

We then fine-tune the whole pipeline in an end-to-end way. The inputs of the diffusion model include the noisy latent z_t, the encoded source image $\mathcal{E}(x)$, the instruction c_{txt}, and the scene priors F_e and F_p from the IAA module. The standard diffusion denoising objective is conditioned on the editing and preserving object priors:

$$\mathcal{L}_{\text{diff}} = \mathbb{E}_{x, c_{\text{txt}}, \epsilon, t}\left[\left\|\epsilon - \epsilon_\theta\left([z_t;\ \mathcal{E}(x)],\ t,\ c_{\text{txt}},\ F_e,\ F_p\right)\right\|_2^2\right], \tag{13}$$

where $[\cdot;\cdot]$ denotes channel-wise concatenation. $\epsilon \sim \mathcal{N}(0, I)$ is the Gaussian noise injected in the forward diffusion process. ϵ_θ denotes the noise prediction network parameterized by θ.

4 Experiments

4.1 Experimental Setup

Implementation Details. GPT-4o is used as the pretrained MLLM. We adopted Stable Diffusion XL [36] as the base model, using 50 sampling steps and a guidance scale of 7.5. To ensure fair comparisons, all competing methods were evaluated using their recommended hyperparameters during inference. All experiments are conducted on NVIDIA A100 GPUs. In the first training stage, the ALP module is pretrained on COCOA [56] for 20,000 iterations. We optimize the ALP module using AdamW with an initial learning rate of 1×10^{-4} and cosine learning rate decay. In the second training stage, we jointly fine-tune the whole pipeline for 20,000 iterations with a batch size of 1. We use AdamW with a weight decay of 0.01 and a short warmup schedule to stabilize optimization.

Evaluation Metrics. Following prior works [48], we evaluate instruction-based local image editing from both semantic and visual aspects. For EMU-Edit [40], we report CLIPim and CLIPout to measure semantic alignment between the edited image and the instruction, and DINO to assess visual similarity between the edited and original images. For PIE-Bench [17], we report CLIPwhole and CLIPedit to measure instruction-image consistency for the whole image and the edited region, respectively, along with DINO for visual fidelity. Since CLIP-based metrics may not fully capture editing quality (e.g., incomplete edits or unintended modifications), we also adopt the evaluation protocol from OmniEdit [45], which uses GPT-based as an automated judge to provide key scores on a scale of 0-10. Specifically, GPT-4o serves as an automatic judge to produce GPT-Success (i.e., instruction adherence) and GPT-Overedit (i.e., preservation of unedited regions) scores. The reported GPTscore is computed as the harmonic mean of the two scores.

Compared Methods. To thoroughly evaluate the performance of our proposed *AmodalEdit*, we compare it with a set of representative image editing methods, including UltraEdit [54], SmartEdit [13], AnyEdit [48], SmartFreeEdit [41], ICEdit [53], and IEAP [12]. We also add large-scale multimodal foundation models for comparison, including FLUX.1 Kontext [19], and Qwen-Image-Edit [46].

4.2 Benchmark Construction

We first conduct evaluations on two widely used instruction-based image editing benchmarks, i.e., EMU-Edit [40] and PIE-Bench [17]. EMU-Edit [40] contains 3,589 samples with 8 distinct editing types under diverse scenarios. PIE-Bench [17] contains 700 samples with 10 editing types, offering fine-grained annotations for assessing both global image quality and local editing accuracy.

In addition, as the above benchmarks only contain a limited number of instructions for occlusion scenarios, we further construct a new benchmark, i.e., OIE-Bench, as follows. OIE-Bench is built upon the object-layered annotations of MULAN [44], which provide explicit object decomposition and relative ordering cues for identifying object occlusions. We select multi-object scenes with obvious occlusion patterns and manually design instructions that either edit occluded objects or alter the original occlusion relationships. The instructions cover common operations, including object addition, movement, replacement, attribute change, removal, and so on.

For each image-instruction pair, GPT-Image-1.5 [34] generates 4 candidate results. The two lowest-quality candidates are filtered out automatically, and the remaining two are independently reviewed by three annotators. A candidate is accepted only when approved by at least two annotators, after which the selected reference is further refined via manual post-processing. Consequently, OIE-Bench provides explicitly occlusion-targeted instructions and carefully curated reference images for more complex scenarios that have not been addressed before.

4.3 Results

Quantitative Evaluation. The quantitative results on public benchmarks are summarized in Table 1. Although these scenarios are not specifically designed for occlusion-aware evaluation, many cases still involve partial overlapping issues, where effective occlusion reasoning benefits semantic alignment and visual consistency.

Method	Size	PIE-Bench				EMU-Edit				Time
		CLIPwhole ↑	CLIPedit↑	DINO↑	GPTscore ↑	CLIPim ↑	CLIPout↑	DINO↑	GPTscore↑	
UltraEdit [54]	7.7B	0.221	0.206	0.794	3.988	0.865	0.254	0.793	3.776	**3.4s**
SmartEdit [13]	14.1B	0.236	0.214	0.814	4.685	0.841	0.278	0.807	4.345	6.4s
AnyEdit [48]	2.9B	0.231	0.204	0.801	4.822	0.862	0.259	0.815	4.683	3.5s
SmartFreeEdit [41]	**1.99B**	0.246	0.221	0.827	5.344	0.855	0.274	0.834	5.711	25s
ICEdit [53]	17B	0.238	0.219	0.854	5.820	0.882	0.271	0.846	5.037	8.6s
IEAP [12]	8.3B	0.235	0.228	0.843	6.241	0.874	0.268	0.827	6.188	–
Ours	4.3B	**0.249**	**0.233**	**0.863**	**7.536**	**0.903**	**0.281**	**0.856**	**6.738**	4.3s
FLUX.1 Kontext [19]	12B	0.247	0.231	0.853	7.298	0.893	0.283	0.862	7.030	31.2s
Qwen-Image-Edit [46]	20B	**0.257**	**0.247**	**0.868**	**8.567**	0.901	**0.286**	**0.868**	**7.940**	62.7s

Table 1: Quantitative comparison results on PIE-Bench [17] and EMU-Edit [40] datasets. The best and second-best results are marked in bold and underlined. We also list the results of open-sourced large-scale multimodal foundation models.

Model	CLIPim↑	CLIPout↑	DINO↑	GPTscore↑
UltraEdit [54]	0.785	0.261	0.770	3.676
SmartEdit [13]	0.827	0.261	0.822	3.211
AnyEdit [48]	0.835	0.252	0.819	5.393
SmartFreeEdit [41]	0.853	0.257	0.844	5.097
ICEdit [53]	0.849	0.241	0.837	5.771
IEAP [12]	0.796	0.263	0.785	5.105
Ours	**0.880**	**0.264**	**0.864**	**7.236**
FLUX.1 Kontext [19]	0.879	0.253	**0.846**	7.165
Qwen-Image-Edit [46]	**0.880**	**0.271**	0.828	**8.104**

Table 2: Quantitative comparison results on our proposed OIE-Bench dataset. The best and second-best results are marked in bold and underlined.

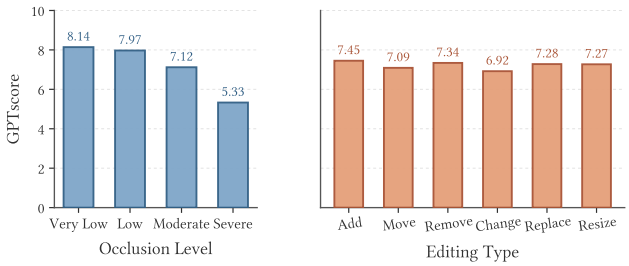


Figure 3: Systematic evaluation across different occlusion levels and editing types, including GPTscore under different occlusion levels (left), and GPTscore with different editing types (right).

Overall, our method achieves the strongest performance on EMU-Edit [40] and PIE-Bench [17]. Our model achieves the highest CLIPim score (0.903) and DINO score (0.856) on EMU-Edit [40], demonstrating that our model maintains strong image-level consistency and structural integrity. Our method also achieves the best results across all reported metrics on PIE-Bench, with particularly strong DINO and GPTscore performance. These results indicate that our occlusion-aware design supports both accurate local editing and coherent integration with the surrounding context. Compared with open-source large-scale multimodal models such as FLUX.1 Kontext [19] (12B) and Qwen-Image-Edit [46] (20B), our model remains competitive on key visual metrics while offering advantages in model size and inference efficiency, achieving a favorable balance between editing quality and deployment cost.

We also evaluate the editing performance on our proposed OIE-Bench. Image editing in these occlusion scenarios requires not only proper target modification, but also reliable handling of overlap boundaries and newly revealed regions. As shown in Table 2, our method consistently surpasses existing methods. The gains in CLIPim and DINO indicate better visual consistency and structural preservation, while the highest CLIPout score suggests that our method is more effective at intended target manipulation without disturbing overlapping objects or surrounding context. Note that our results are even better than those of large-scale foundation models for the DINO score, further demonstrating the benefits of amodal scene layout perception in instruction-based image editing.

We further conduct a systematic analysis of model performance across different occlusion levels and editing types on OIE-Bench. As shown in Figure 3, the GPTscore gradually decreases from 8.14 under very low occlusion to 5.33 under severe occlusion, indicating that heavily occluded scenes remain substantially more challenging due to the increased ambiguity in hidden-region reconstruction. Nevertheless, our method maintains stable performance across different editing operations, with GPTscore ranging from 6.92 to 7.45. These results suggest that the model generalizes well across diverse editing types, while its performance is more sensitive to occlusion severity than to the specific editing operation.

Qualitative Evaluation. Figure 4 provides qualitative comparisons between our model and existing methods under complex scenarios. UltraEdit [54] and SmartEdit [13] tend to be less responsive to edits involving overlapping objects, as shown in the chair insertion and shoe movement cases. While AnyEdit [48] demonstrates better instruction adherence, it occasionally struggles to disentangle the target object from its occluder, leading to structural ambiguities or less coherent reconstruction of newly exposed regions. SmartFreeEdit [41], IEAP [12], and ICEdit [53] show strong editing capabilities, yet hidden-region recovery remains challenging when occlusion relationships change, resulting in texture inconsistencies, incomplete completion, or integration artifacts.

Different from existing works, our method mitigates these challenges by explicitly modeling occlusion relationships and reasoning

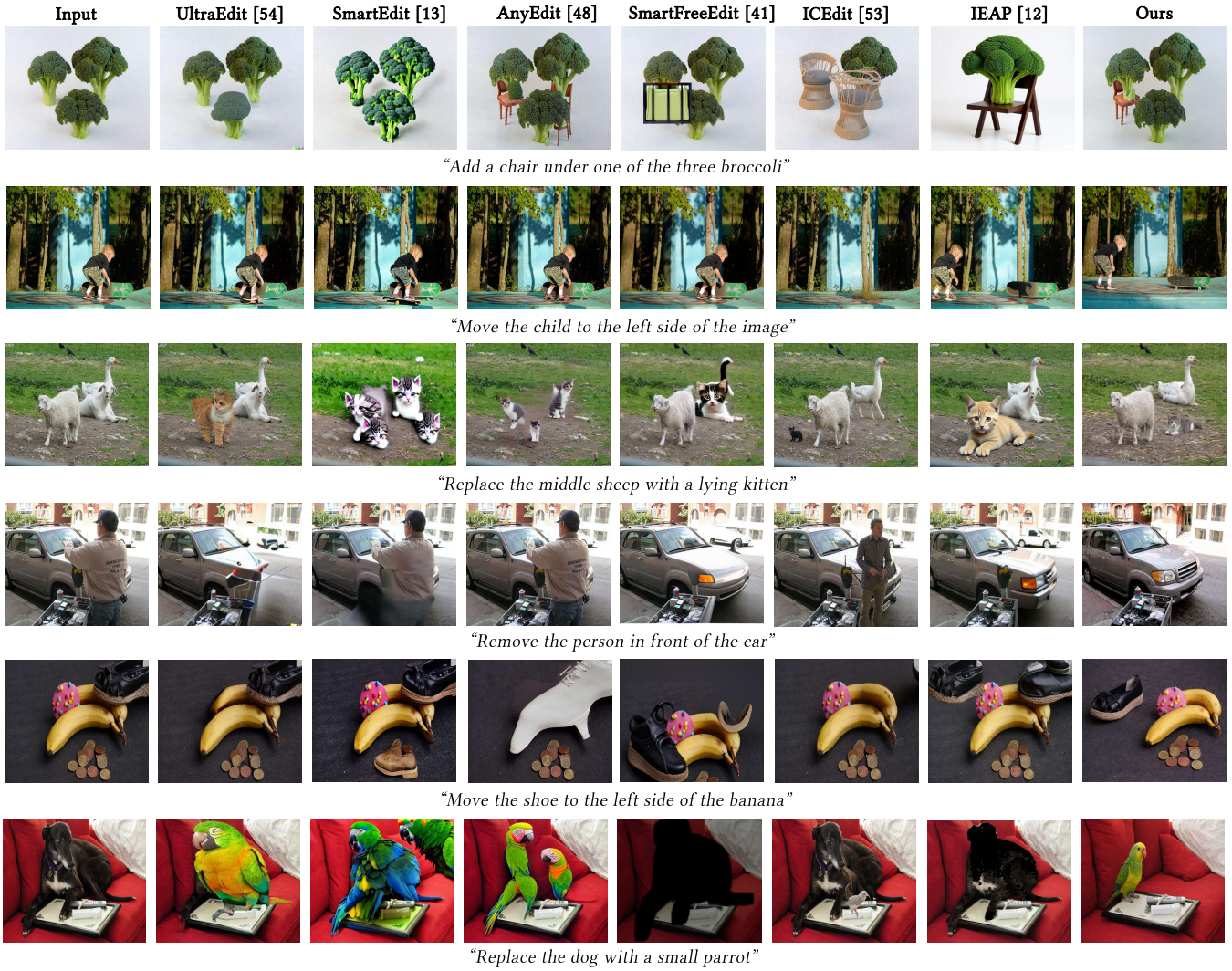


Figure 4: Qualitative comparison. Given an input image (1*st* column) and the corresponding text instruction (below each row), we show the image editing results from different methods (2*nd* to 7*th* columns) and our model (8*th* column).

Model	CLIPim↑	CLIPout↑	DINO↑	GPTscore↑
w/o ALP	0.774	0.242	0.783	3.124
w/o ET	0.866	0.258	0.844	6.480
w/o PT	0.818	0.248	0.824	4.967
Ours (full)	**0.880**	**0.264**	**0.864**	**7.236**

Table 3: Results of the ablation study. The best results are marked in bold.

about the invisible parts affected by editing. Our model correctly places the chair behind the broccoli, restores the skateboard after moving the child, recovers the grass during sheep replacement without affecting nearby objects, and reconstructs the car after person removal. These results demonstrate that our method achieves more reliable object disentanglement, hidden-region recovery, and structural preservation than existing baselines.

4.4 Ablation Study

To understand how each design choice in our model contributes to instruction-based image editing, we conduct ablation studies by considering the following variants and analyze the robustness of our method to different external perception modules.

Effect of Amodal Layout Perception (i.e., w/o ALP). Removing ALP consistently degrades all metrics in Table 3. Without amodal structural guidance, the model relies more on generative priors to reconstruct newly exposed regions, resulting in less reliable geometry. This demonstrates the importance of amodal cues for coherent structural completion.

Graph Setting	CLIPim↑	CLIPout↑	DINO↑	GPTscore↑
Oracle graph	**0.887**	**0.267**	**0.870**	**7.562**
Predicted graph	0.880	0.264	0.864	7.236
Noisy predicted graph	0.843	0.248	0.837	5.792
No occlusion graph	0.795	0.247	0.787	4.229

Table 4: Effect of occlusion graph quality. The best results are marked in bold.

MLLM	Object Detector	CLIPim↑	CLIPout↑	DINO↑	GPTscore↑
GPT-4o [33]	MaskDINO [20]	**0.880**	0.264	**0.864**	**7.236**
Qwen2.5-VL-32B [2]	MaskDINO [20]	0.866	**0.266**	0.859	6.983
Qwen2.5-VL-7B [2]	MaskDINO [20]	0.851	0.258	0.835	6.641
Qwen2.5-VL-7B [2]	YOLOv8 [16]	0.851	0.261	0.829	6.604
GPT-4o [33]	YOLOv8 [16]	0.874	0.263	0.857	7.182

Table 5: Robustness to different external MLLMs and object detectors. The best results are marked in bold.

Effect of Editing Token Injection (w/o ET). Without editing token injection, the editing branch receives only the text instruction and lacks direct structural cues about the original target. As shown in Table 3, editing tokens improve object-level transformations by providing priors about the target's complete structure, leading to more consistent and geometrically plausible results.

Effect of Preserving Token Injection (w/o PT). We remove the preserving pathway by redirecting its tokens to the editing branch. Since amodal tokens encode the original scene structure, injecting them solely into the editing branch introduces a strong bias toward the original content. The results in Table 3 demonstrate that the preserving branch is important for decoupling structural preservation from semantic manipulation.

Effect of Occlusion Graph Quality. We evaluate the sensitivity of our method to the quality of the occlusion graph. In addition to the predicted graph used in our full model, we consider an oracle graph, a noisy predicted graph, and a variant without occlusion relations. For the noisy graph, we perturb 30% of the edges by randomly deleting existing edges and adding incorrect ones. As shown in Table 4, the predicted graph achieves performance close to that of the oracle graph, while introducing noisy edges causes consistent degradation across all metrics. Removing the occlusion graph results in a more substantial performance drop, demonstrating that explicit occlusion relations provide essential structural cues for amodal reasoning and editing.

Robustness to External Perception Modules. We further examine whether our method depends heavily on external MLLMs or object detectors. Specifically, we replace GPT-4o [33] and MaskDINO [20] with different Qwen2.5-VL variants [2] and YOLOv8 [16], respectively. As shown in Table 5, our method retains strong performance across different configurations, with only moderate variations in the evaluation metrics. These results demonstrate that the proposed pipeline is robust to the prediction noise introduced by different external modules.

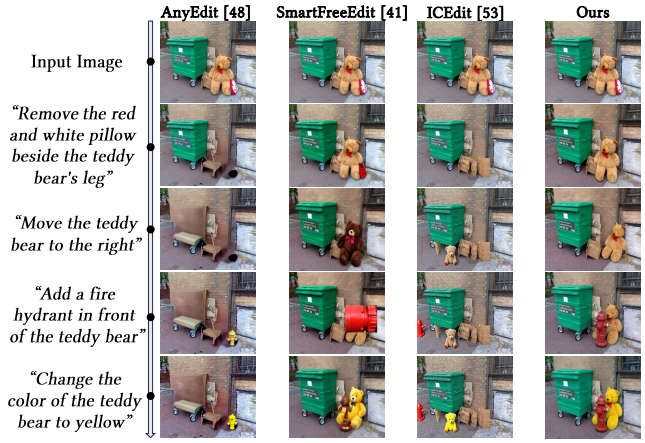


Figure 5: Results of multi-turn image editing. Given an input image (1*st* row), we show the sequential editing results (2*nd* to 5*th* rows) of existing methods (1*st* to 3*rd* columns) and our model (4*th* column).

4.5 Multi-Turn Image Editing

Multi-turn image editing aims to manipulate an input image through a sequential trajectory of modifications, while necessitating the preservation of global scene coherence at each editing step. By leveraging amodal scene priors, *AmodalEdit* is well suited to both single-turn and multi-turn editing.

As shown in Figure 5, we compare our method with existing approaches using four-step editing sequences. Existing methods exhibit different forms of accumulated error. In the left example, AnyEdit [48] incorrectly removes both the pillow and the teddy bear in the first turn, preventing subsequent edits from being properly executed. SmartFreeEdit [41] better preserves the initial layout, but gradually introduces appearance and scale inconsistencies, including an oversized fire hydrant that disrupts the scene composition. ICEdit [53] also suffers from identity drift over multiple turns.

In contrast, our method accurately removes the target, reconstructs the cardboard boxes revealed after moving the teddy bear, places the fire hydrant with a plausible occlusion relationship, and keeps the final attribute edit localized. These results further demonstrate the importance of amodal scene layout perception for instruction-based image editing across diverse scenarios.

5 Conclusion

In this paper, we present *AmodalEdit*, a new framework for instruction-based image editing via amodal scene layout perception. Unlike existing methods that rely on the explicit visibility of target objects, our method leverages amodal scene understanding priors to advance instruction-based image editing in diverse scenarios with object occlusions and interactions. Extensive experiments on a variety of benchmarks demonstrate that our method outperforms existing methods across multiple metrics. Furthermore, we showcase the superiority of *AmodalEdit* in multi-turn image editing by maintaining the amodal scene layout across editing sequences.

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